VHDL NOT Gate Simulation Report

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Objective: Implement a NOT gate in VHDL, connect the VHDL entity in Logisim-evolution, and verify that the simulated output matches the NOT-gate truth table.

1. VHDL Code

```
library ieee;
use ieee.std_logic_1164.all;

entity not_gate is
    port (
        a : in std_logic;
        y : out std_logic
    );
end not_gate;

architecture behavioral of not_gate is
begin
    y <= not a;
end behavioral;
```

2. Expected Operation

Input a	Output y
0	1
1	0

A NOT gate inverts its input. Therefore, when a = 0 the expected output is y = 1, and when a = 1 the expected output is y = 0.

3. Logisim-evolution Simulation Results

Case 1: a = 0, y = 1

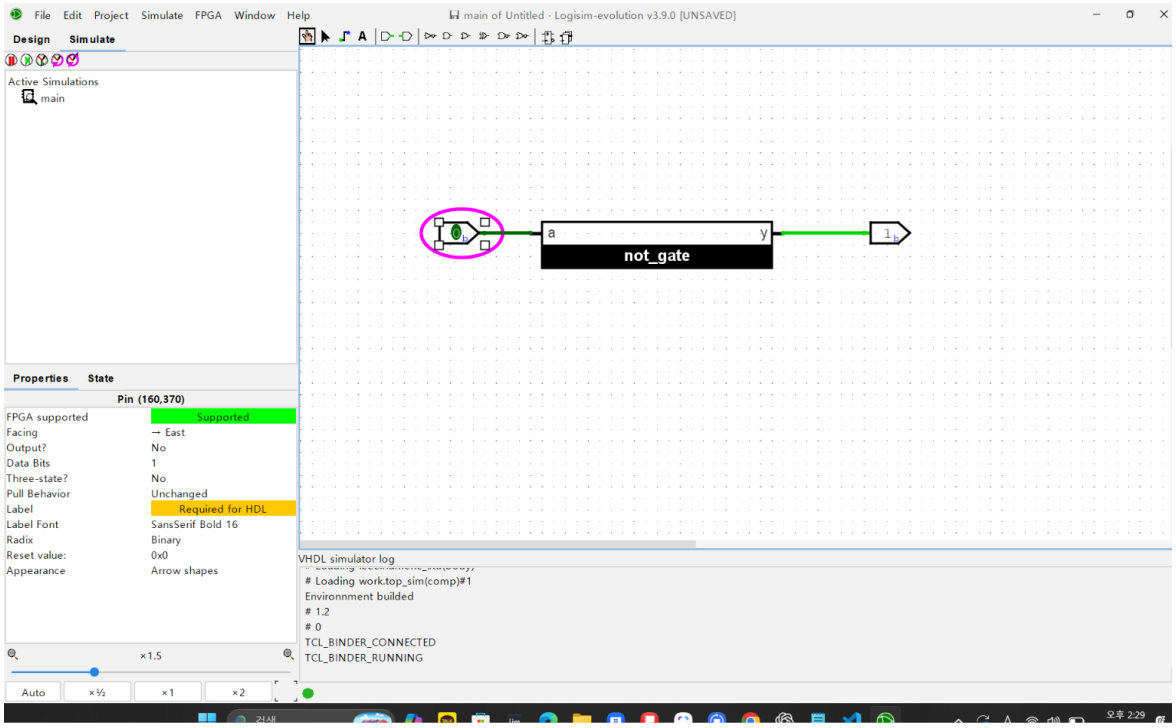


Figure 1. The input is 0 and the VHDL NOT gate produces output 1.

Case 2: a = 1, y = 0

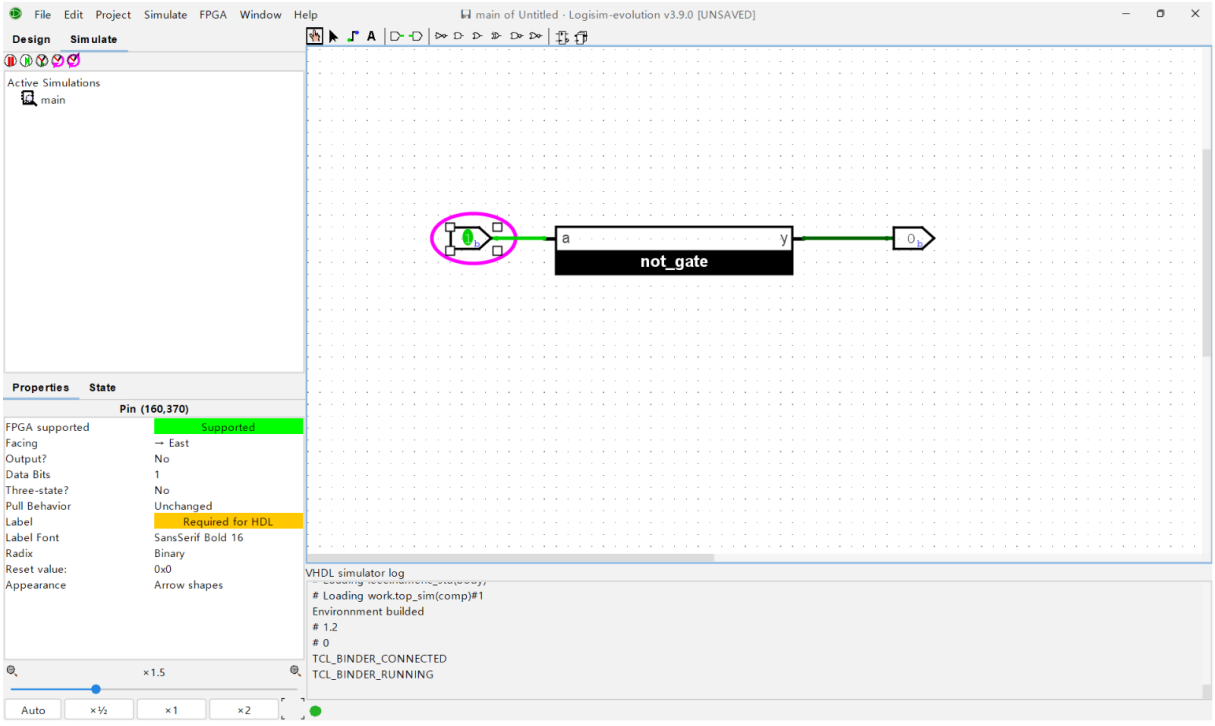


Figure 2. The input is 1 and the VHDL NOT gate produces output 0.

4. Result and Conclusion

The Logisim-evolution simulation produced y = 1 when a = 0 and y = 0 when a = 1. Both results match the expected truth table of a NOT gate. Therefore, the VHDL NOT-gate implementation and the Logisim-evolution simulation were verified successfully.

Files to submit to GitHub

• VHDL source code (.vhd)
• Logisim project (.circ)
• PDF report